

Here’s a **complete, structured list of internal and external cyber security threats**, along with how **Artificial Intelligence (AI)** can help detect, prevent, and mitigate them effectively.

☐ **CYBER SECURITY THREATS — INTERNAL & EXTERNAL**

Category	Type of Threat	Description	AI-based Handling / Mitigation Technique
☐ Internal Threats	Insider Threats (Malicious Employee)	Employees intentionally leaking, stealing, or sabotaging sensitive data.	AI monitors user behavior patterns (User Behavior Analytics - UBA) and flags anomalies in access or data movement.
	Accidental Data Leakage	Employees unintentionally share confidential data via email or cloud.	AI email scanners and NLP models detect sensitive data (e.g., PII) before sending messages.
	Privilege Misuse	Overuse or abuse of administrative privileges by insiders.	AI-based access management and anomaly detection systems restrict suspicious privilege escalations.
	Shadow IT	Employees using unauthorized apps or services for work.	AI tracks network traffic and automatically identifies unapproved applications.
	Negligence or Poor Security Hygiene	Weak passwords, not following policies, or ignoring updates.	AI-driven security assistants remind users, enforce policies, and block risky behaviors.
	Phishing Response by Employees	Employees clicking phishing links.	AI phishing simulation tools and adaptive training systems personalize awareness programs.

Category	Type of Threat	Description	AI-based Handling / Mitigation Technique
☐ External Threats	Phishing Attacks	Fake emails or websites to steal credentials.	AI-powered email filters and NLP models detect phishing intent in real-time.
	Malware & Ransomware	Malicious software that encrypts or steals data.	AI uses pattern recognition and behavior-based detection to identify and quarantine malware before

			execution.
	DDoS Attacks (Distributed Denial of Service)	Attackers flood servers with fake requests to crash systems.	AI models predict abnormal traffic surges and auto-scale firewalls or redirect traffic.
	Zero-Day Exploits	Attacks exploiting unknown software vulnerabilities.	AI predicts potential zero-day vulnerabilities by analyzing code and previous exploit patterns.
	Man-in-the-Middle (MitM) Attacks	Intercepting communication between two systems.	AI monitors network traffic anomalies and applies encryption-based adaptive defense.
	SQL Injection / Web Exploits	Attackers inject malicious SQL commands through web inputs.	AI-driven Web Application Firewalls (WAF) detect abnormal query behavior using ML.
	Brute-Force Attacks	Attackers guess passwords repeatedly.	AI detects repetitive failed login attempts and automatically enforces CAPTCHA or lockout.
	IoT Device Exploits	Weakly secured IoT devices becoming entry points.	AI continuously profiles IoT device behavior to isolate compromised nodes.
	Advanced Persistent Threats (APT)	Long-term, stealthy attacks targeting critical assets.	AI threat intelligence systems identify unusual patterns and correlate with known APT signatures.
	Social Engineering	Psychological manipulation to extract confidential info.	AI-driven sentiment and behavior analysis detect manipulation attempts in communication logs.

□ HOW AI HANDLES CYBER SECURITY THREATS (TECHNIQUES)

AI Technique	How It Helps	Examples
Machine Learning (ML)	Learns from historical attack data to detect future anomalies.	Malware detection, intrusion detection systems (IDS).
Deep Learning (DL)	Recognizes complex attack patterns and zero-day exploits.	Image recognition for phishing sites, voice deepfake detection.

Natural Language Processing (NLP)	Understands and classifies text-based threats.	Detecting phishing or malicious intent in emails.
Behavioral Analytics	Tracks user and system behaviors to find irregularities.	Insider threat detection.
Predictive Analytics	Forecasts potential attacks based on trends and data.	Early threat alerts, vulnerability management.
AI-based SIEM (Security Information & Event Management)	Correlates logs and alerts from multiple systems for threat detection.	Splunk, IBM QRadar, Azure Sentinel.
Automated Incident Response (SOAR)	AI automates investigation and response workflows.	Automatically quarantines infected systems or resets credentials.
Generative AI (GenAI)	Generates synthetic threat scenarios for simulation & defense training.	Cyber defense drills and red-team simulations.

❑ AI TOOLS & PLATFORMS USED IN CYBER SECURITY

Tool / Platform	Purpose
Darktrace	AI-based network threat detection and autonomous response.
CrowdStrike Falcon	Endpoint protection using behavioral AI.
CylancePROTECT	Predictive malware prevention using ML.
IBM QRadar	AI-driven SIEM for log and event correlation.
Splunk Enterprise Security	ML-driven real-time monitoring and analytics.
Microsoft Defender with Copilot	AI-assisted threat hunting and response.
Google Chronicle Security	Cloud-based AI analytics for detecting sophisticated threats.

✔ Conclusion

Aspect	AI Role in Cyber Security
Detection	Identifies abnormal activity in real time.
Prevention	Predicts and blocks threats before they happen.
Response	Automates mitigation and recovery actions.
Adaptation	Continuously learns from new attacks.

Would you like me to create a **diagram or architecture flow** showing how **AI handles internal and external cyber threats** (e.g., sensors → AI detection → response system)? It would visually summarize how everything connects.